adapted on
Lynford L. Goddard
Brian T. Cunningham

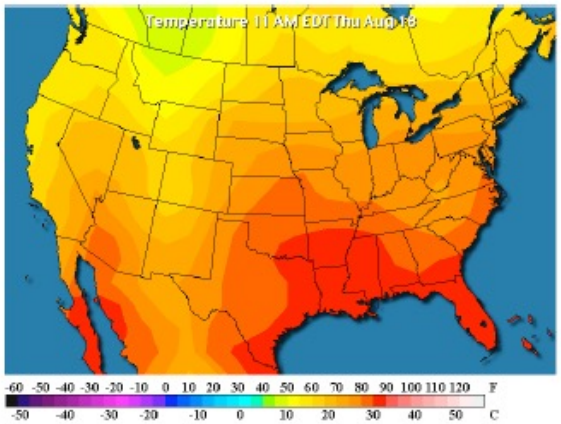
Lectures 2

Scalar and Vector Fields
Lorentz Force Equation
Coulomb's Law

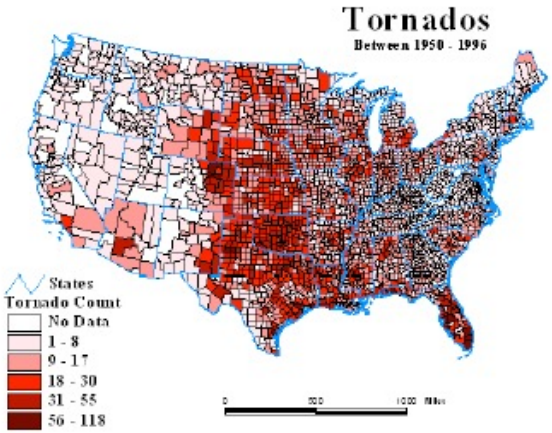
Victoria Shao
yangshao@Illinois.edu

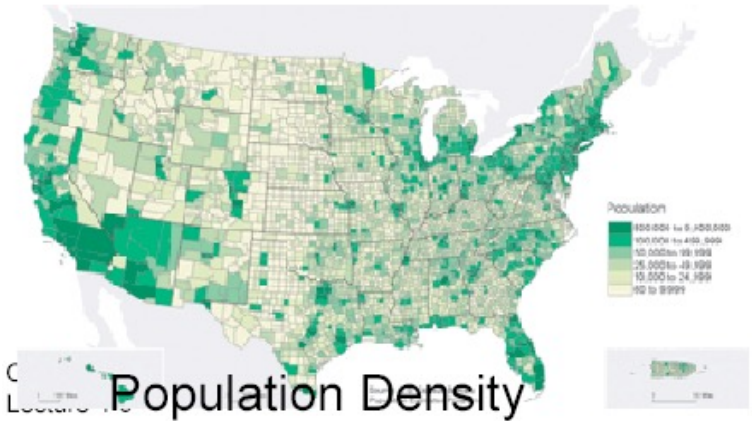
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Scalar Field



Temperature



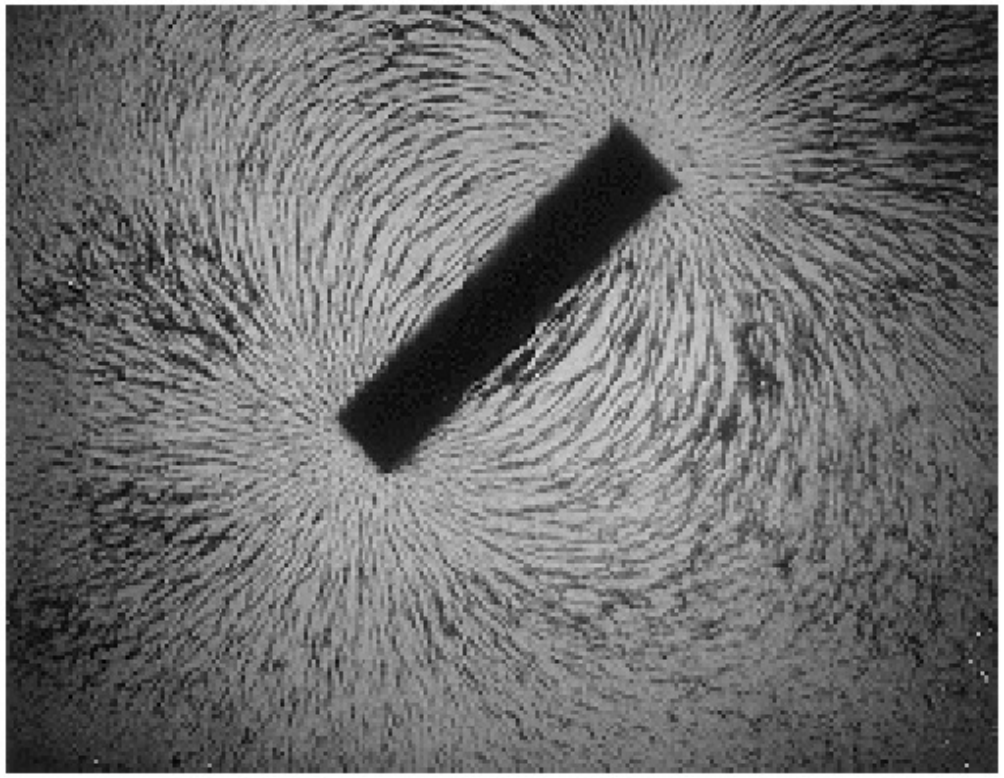

Population Density



Elevation

- length, area, speed,
- volume, mass, density
- pressure
- temperature
- energy, entropy
- work, power
- …

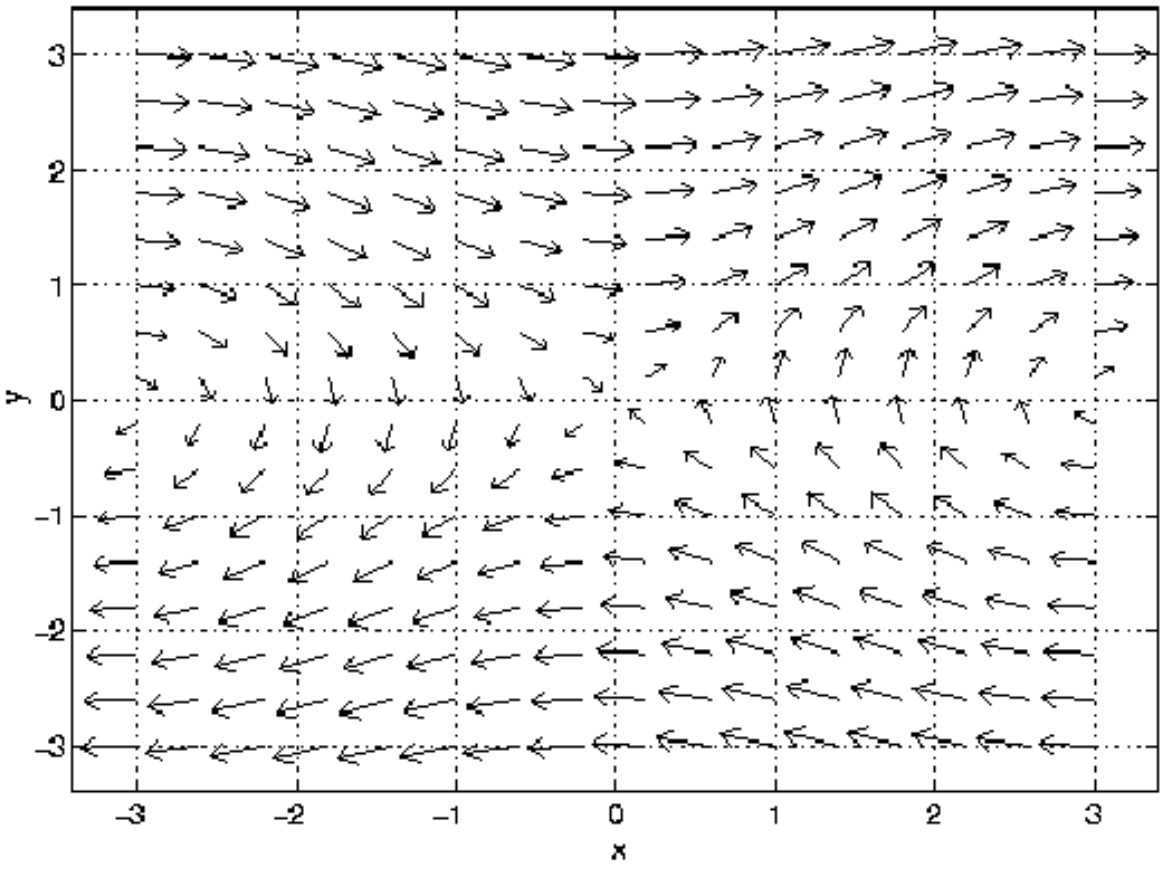
Vector Field



- displacement
- velocity
- acceleration
- momentum
- force
- lift, drag
- …

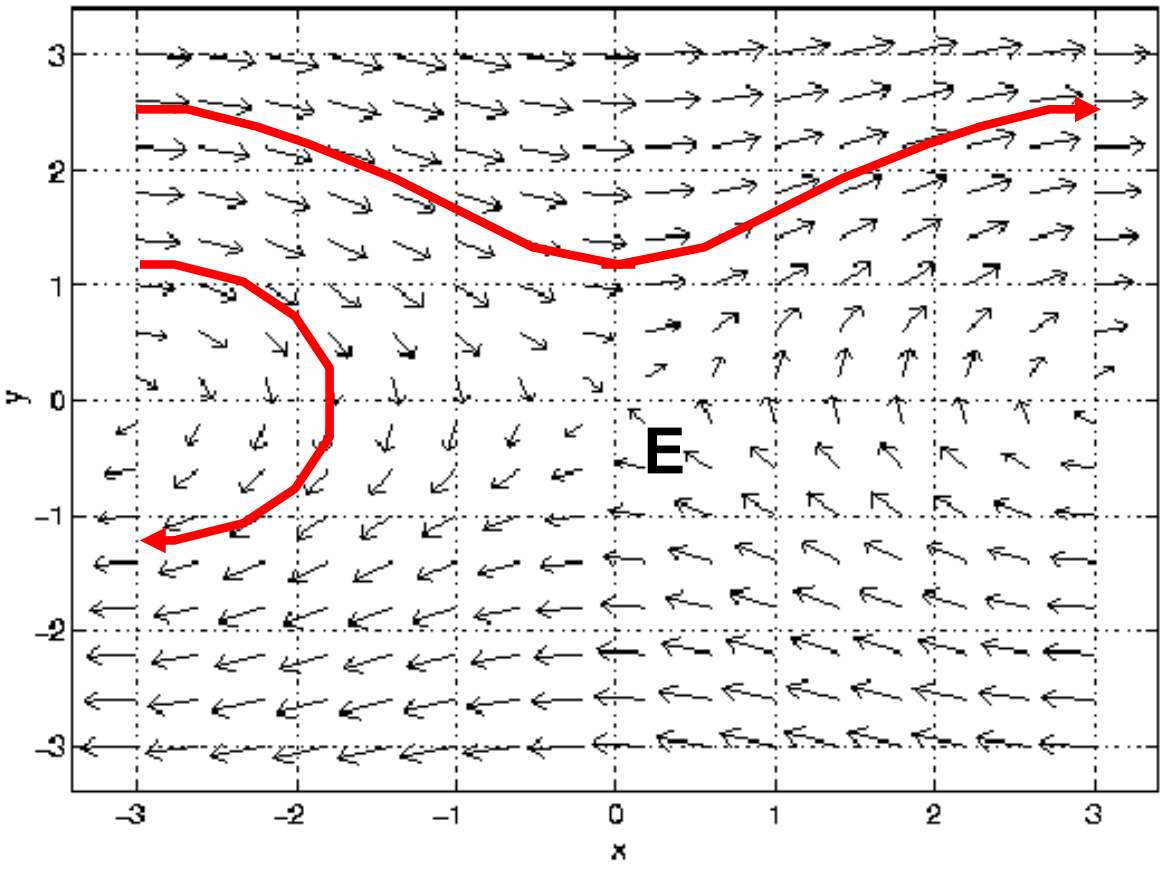
Vector Field

Example $\vec{E}(x,y) = y\,\hat{a}_x + sinx\,\hat{a}_y$



Electric force

$$\vec{F}_E = q\vec{E}$$

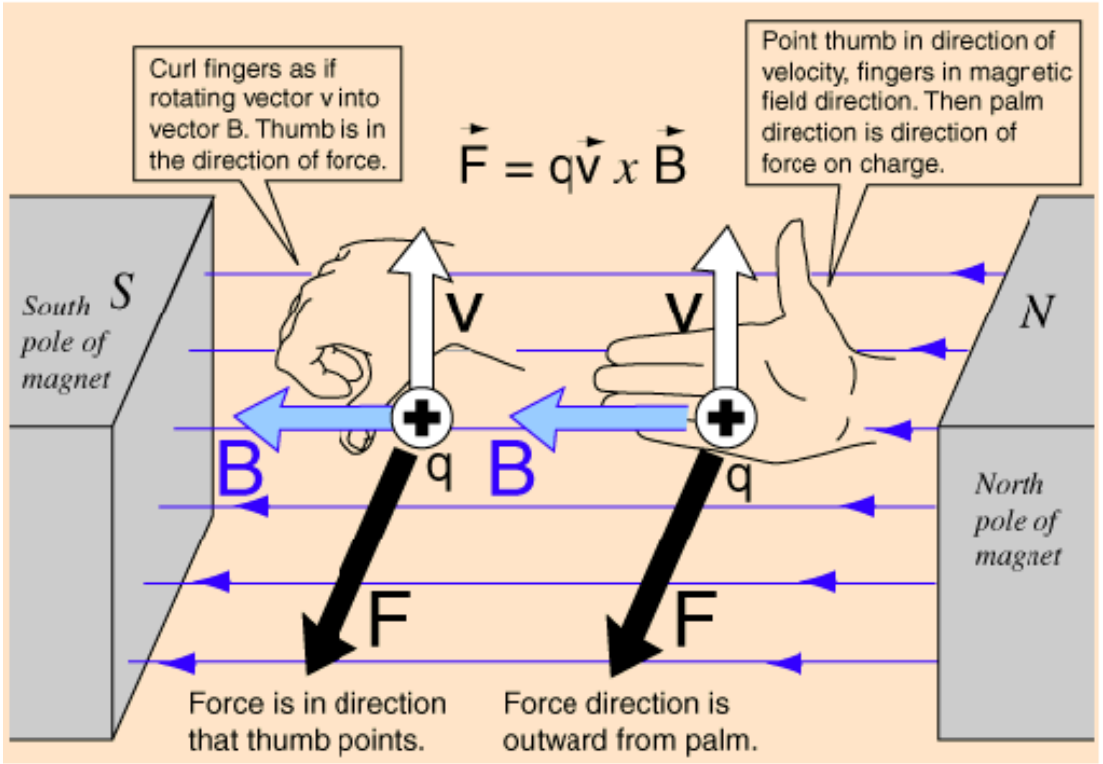


Force vectors $\mathbf{F}_E$ are parallel to **E** for a positive charge q

Magnetic force

$$\vec{F}_M = q\vec{v} \times \vec{B}$$

Direction: right hand rule for positive charge



Lorentz Force Equation

So if a region of space contains BOTH an **E** field and a **B** field, a moving charge will experience force from both at the same time…

$$\vec{F}_{TOTAL} = \vec{F}_E + \vec{F}_M$$

$$\vec{F}_{TOTAL} = q\vec{E} + q\vec{v} \times \vec{B}$$

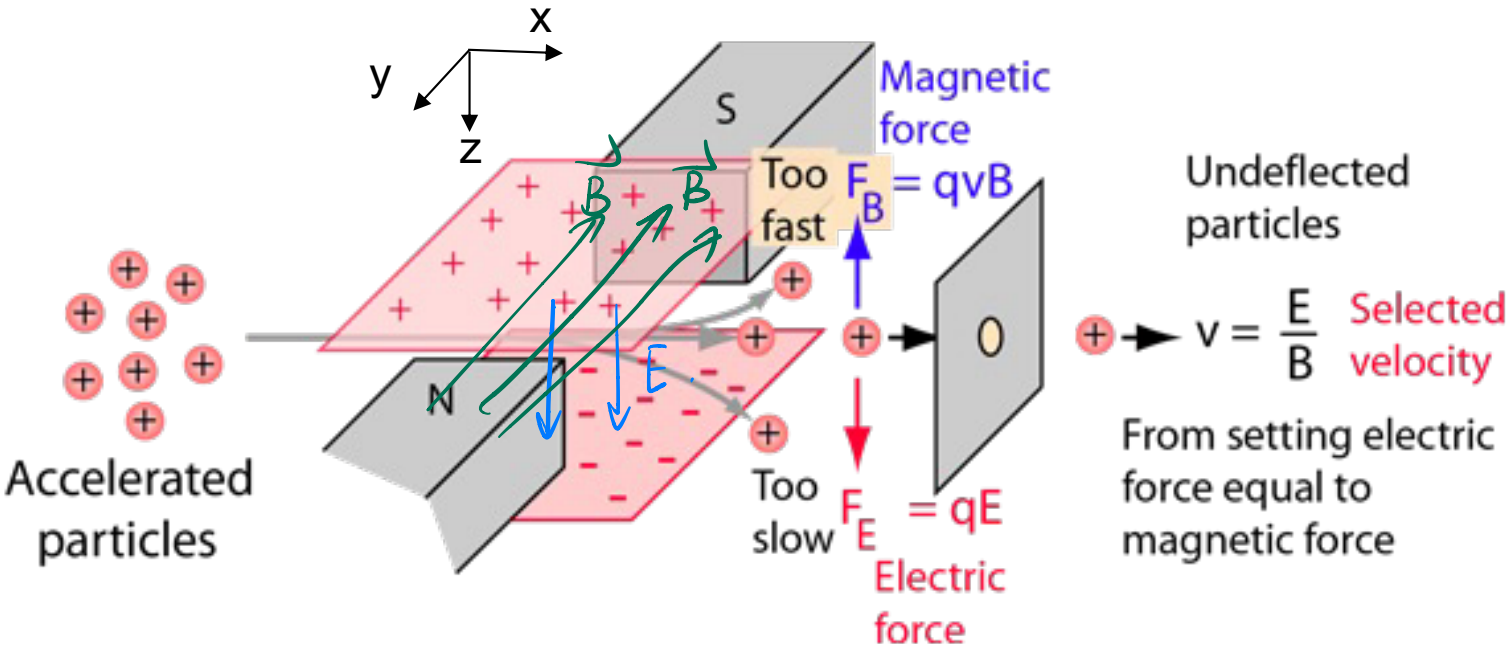
Application: Mass Spectrometers

- Part I: Velocity Selector
  - Particles with a specific velocity in crossed EM fields are undeflected

$$\vec{E} = E_0 \hat{a}_z$$

$$\vec{B} = -B_0 \hat{a}_y$$

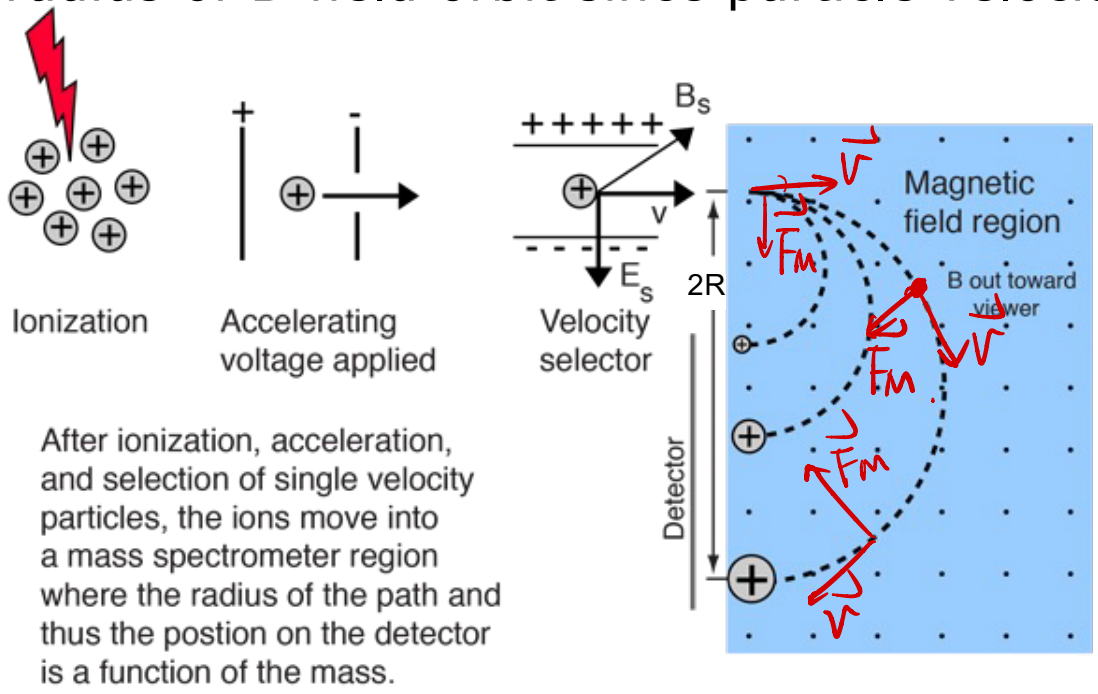
$$\vec{v} = v_0 \hat{a}_x$$



$$\vec{F}_{TOTAL} = q(E_0 - v_0 B_0)\hat{a}_z = 0 \; iff \; v_0 = E_0 / B_0$$

Application: Mass Spectrometers

- Part II: Mass Selector
  - Mass sets the radius of B-field orbit since particle velocity is the same



$$\vec{B} = B_0 \hat{a}_z$$

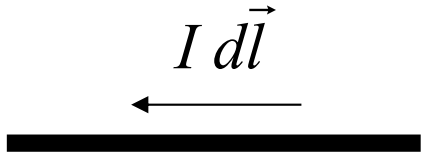
$$\vec{v} = v_0 \hat{a}_x$$

Problem: Solve for R as a function of the particle's mass, m, and velocity, v_0

$$F_c = mv_0^2 / R = qv_0 B_0$$

Current = Moving Charge

What is CURRENT? CHARGES IN MOTION!!



$$I\,d\vec{l} = q\,\vec{v}$$

$$\frac{coul}{\sec}\cdot m = coul\cdot\frac{m}{\sec}$$

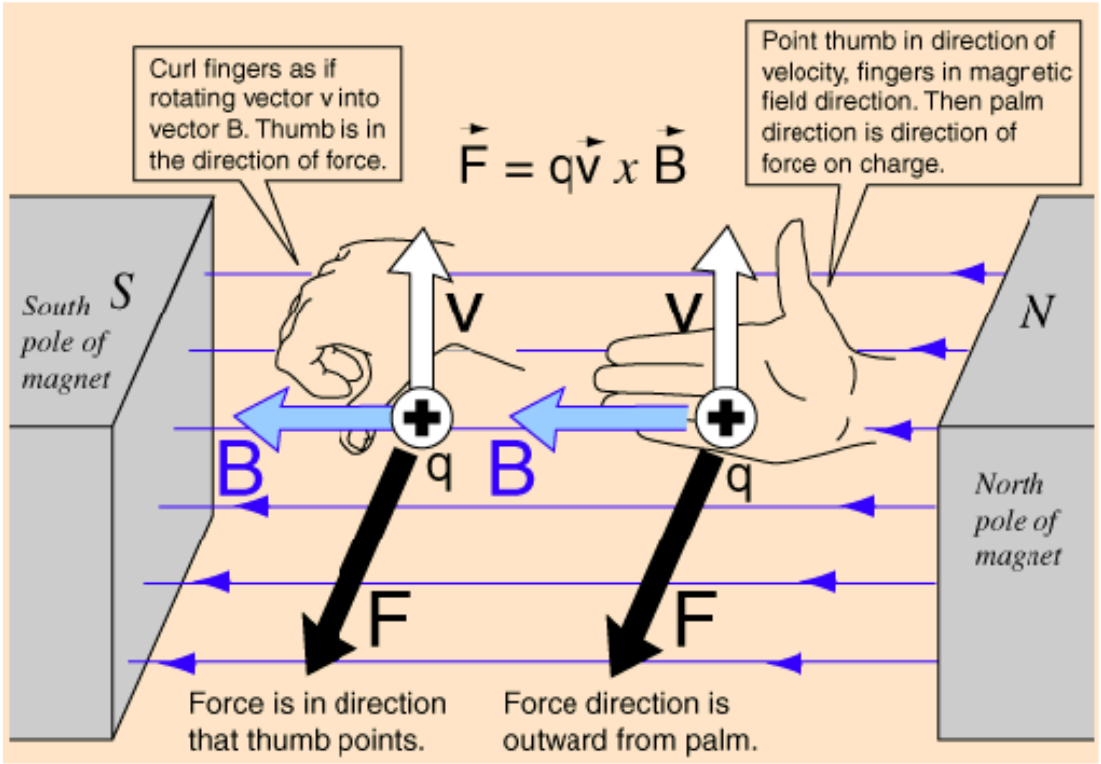
$I = \frac{Q}{\Delta t}$ A.

So the current in a wire, ***I***, flowing across a magnetic field will feel a force…

Magnetic force

$$F_M = (I\,d\vec{l}\,) \times \vec{B} = q\,\vec{v} \times \vec{B}$$

Direction: Perpendicular to velocity vector
Perpendicular to B vector



Electrostatic Force

What is the force F_2 on a point charge Q_2 due to a single point charge Q_1 located a distance R away?



Coulomb's observations

- The magnitude of F is
  - proportional to the product of the charges
  - inversely proportional to the square of the distance
  - depends on the medium
- F points along the joining line
- Like charges repel; unlike charges attract



Coulomb's Law

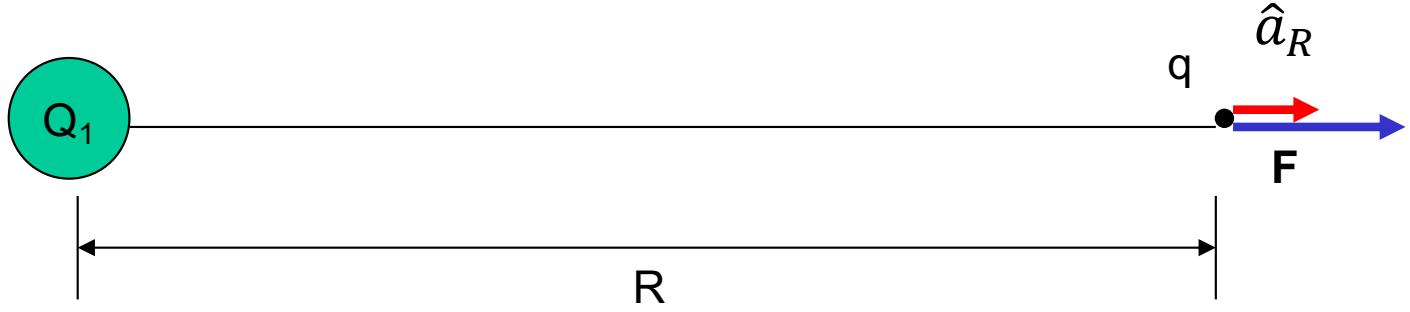


$$\vec{F}_1 = \frac{Q_1 Q_2}{4\pi\varepsilon_0 R^2}\hat{a}_{21} \qquad \vec{F}_2 = \frac{Q_1 Q_2}{4\pi\varepsilon_0 R^2}\hat{a}_{12}$$

Electric Field

- The electric field **E** is the force per unit charge caused by the source charges

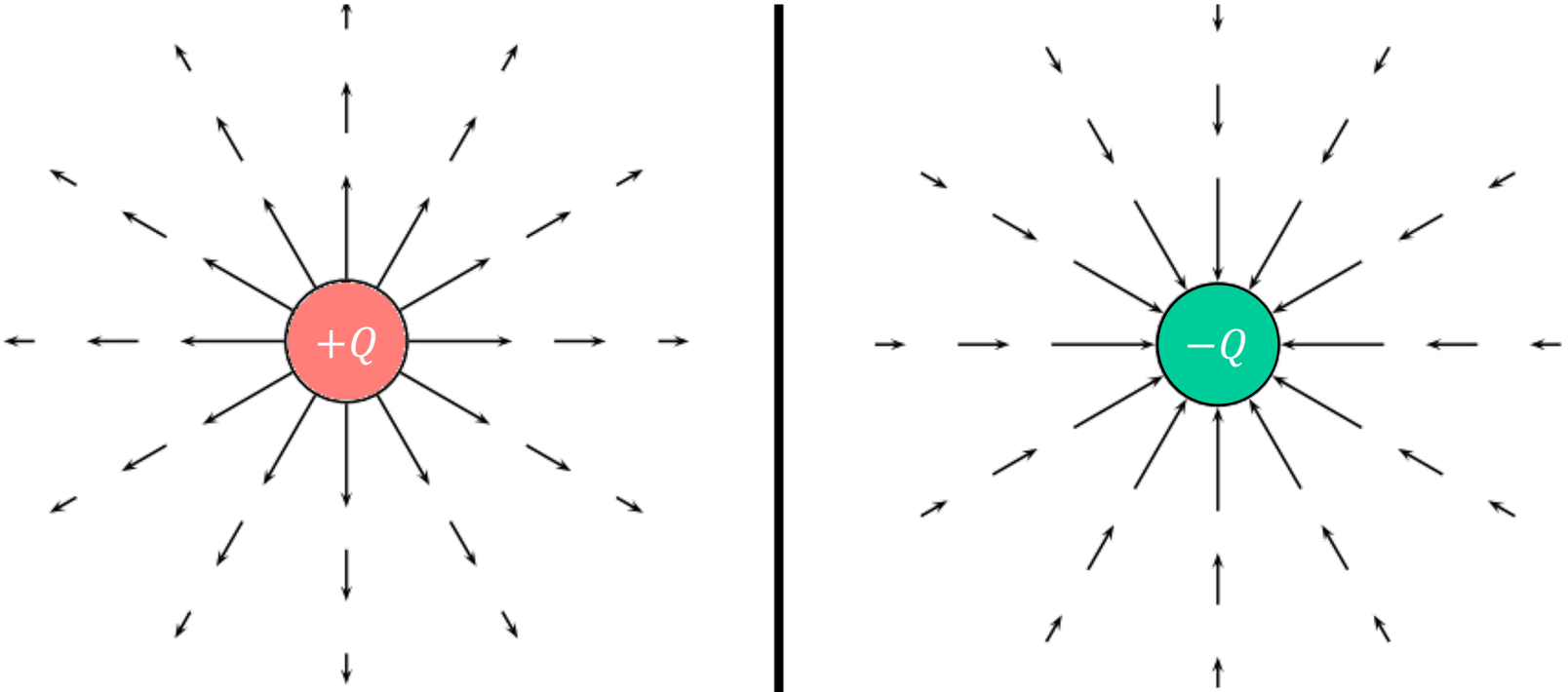
Tiny "test charge" (with + charge)



$\vec{F}_E = q\vec{E}$

$$\vec{E} = \lim_{Q_2 \to 0} \frac{\vec{F}_2}{Q_2} = \frac{Q_1}{4\pi\varepsilon_0 R^2}\hat{a}_R$$

Electric Field Around a Point Charge



Electric field around positive point charge

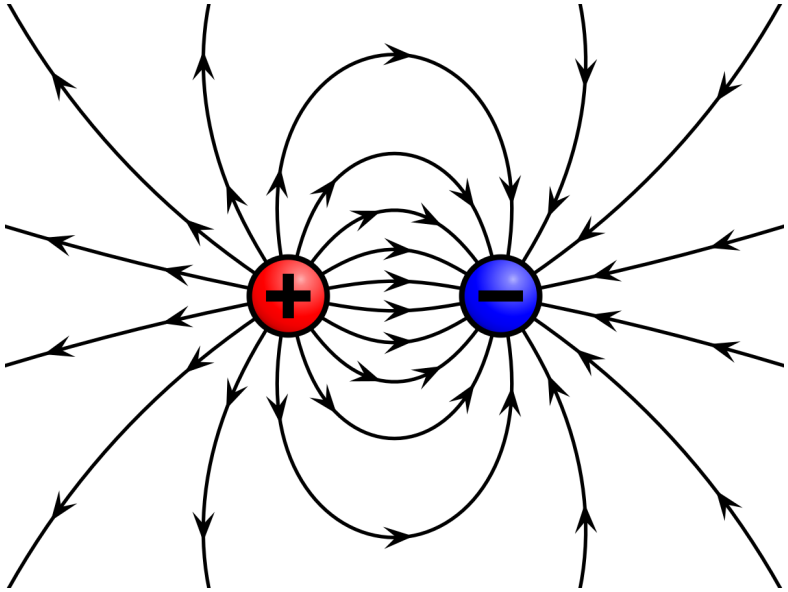
Electric field around negative point charge

Field strength is proportional to the length of vectors

$$\vec{E} = \frac{Q}{4\pi\varepsilon_0 R^2}\ \hat{a}_R$$

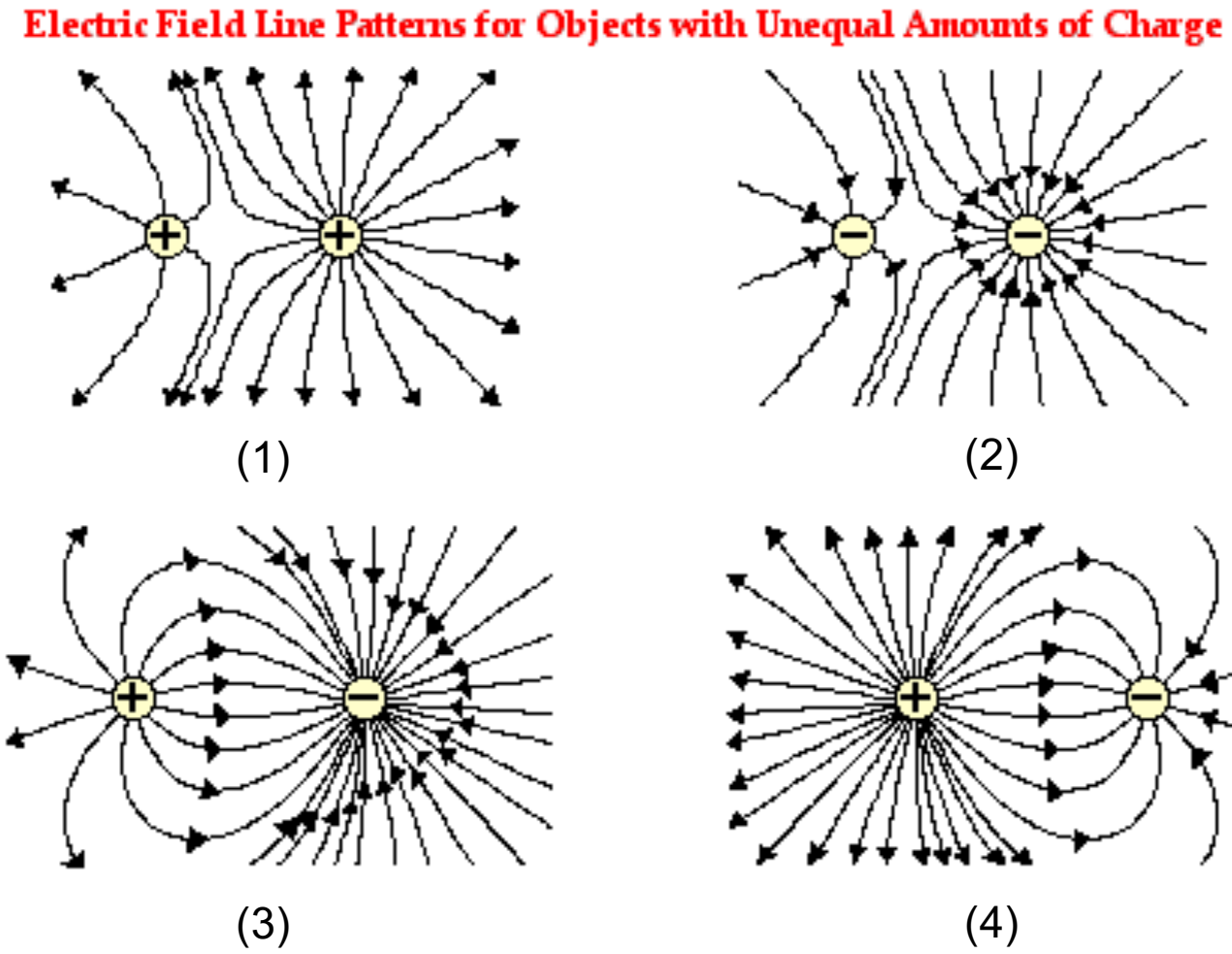
Field Lines

- Another way to graphically represent vector field
- The field strength is proportional to the **density** of field lines
- **E**-field lines begin on + charges, initially emanating uniformly in all directions, and end on – charges
  - Can't stop in midair but can extend to ∞
- They **never** intersect



Electric field of a positive and a negative point charge

Field Lines



Poll: Which source in the last two figures has larger charge Q?

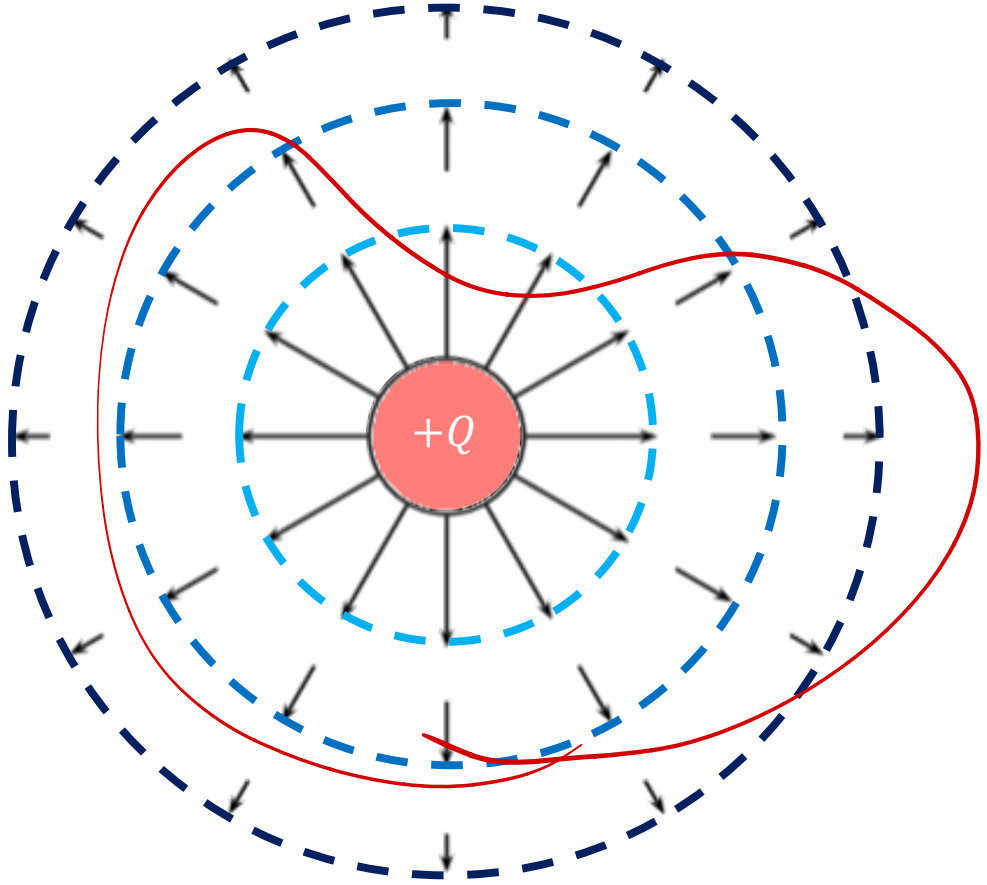
A) + charge in (3),
+ charge in (4)

B) + charge in (3),
- charge in (4)

C) - charge in (3),
+ charge in (4)

D) - charge in (3),
- charge in (4)

Surfaces of Constant E Magnitude are Spheres

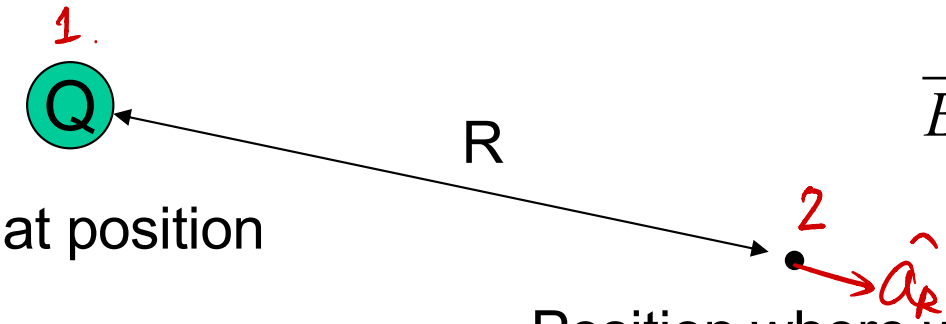


Electric Field Around a Positive Point Charge

$$\vec{E} = \frac{Q}{4\pi\varepsilon_0 R^2}\ \hat{a}_R$$

Gauss' Law
Number of field lines passing thru **any** surface that encloses Q_1 is constant

Calculating the Electric Field



$$\vec{E} = \frac{Q}{4\pi\varepsilon_0 R^2}\,\hat{a}_R$$

Point Charge at position (x_1,y_1,z_1)

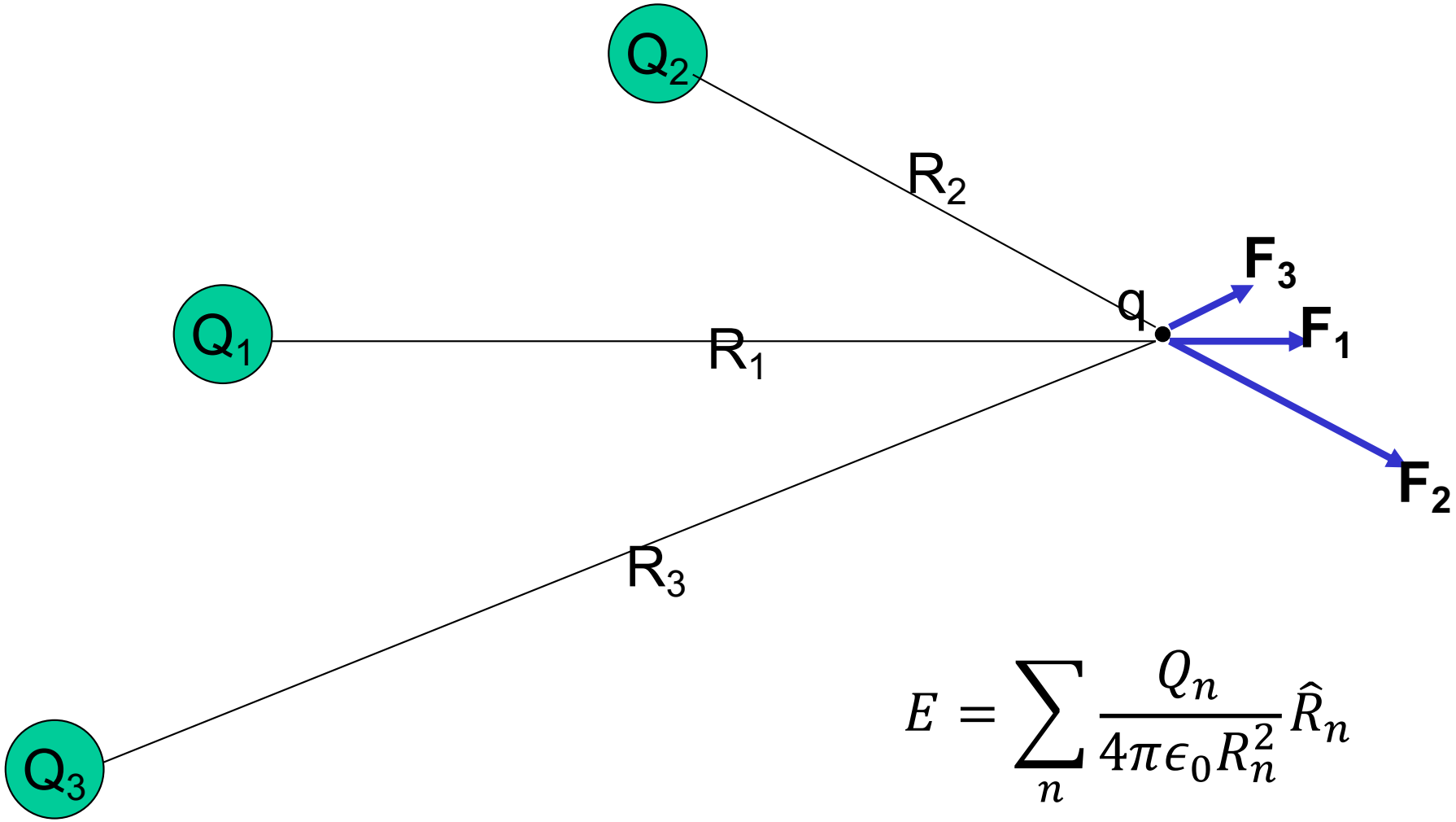
Position where we want to calculate electric field at Position (x_2,y_2,z_2)

$$R = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

$$\hat{a}_R = \frac{(x_2 - x_1)\hat{a}_x + (y_2 - y_1)\hat{a}_y + (z_2 - z_1)\hat{a}_z}{R}$$

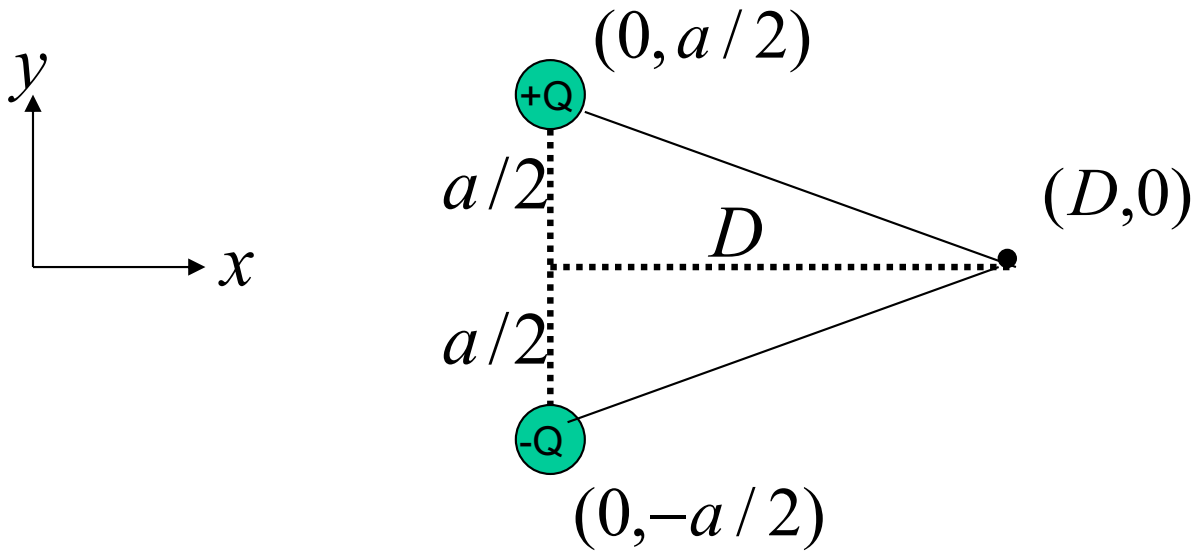
Unit vector pointing along direction from Q to Point

Superposition of E Fields



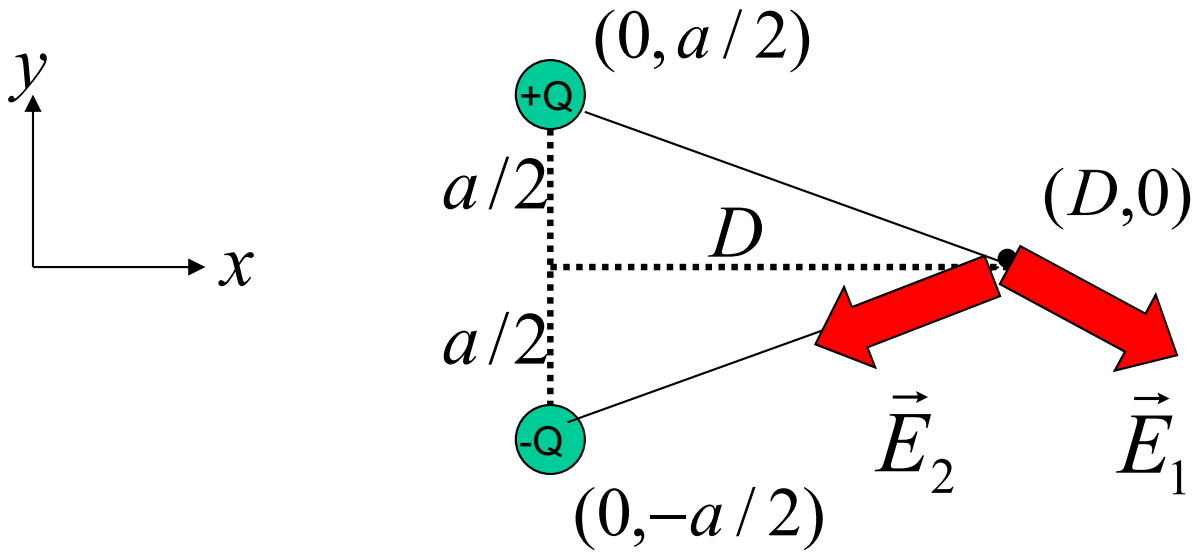
$$E = \sum_n \frac{Q_n}{4\pi\epsilon_0 R_n^2} \hat{R}_n$$

Example: Calculate the E-field of a Dipole at a Point



Tip: We only need to work in 2D. Why?

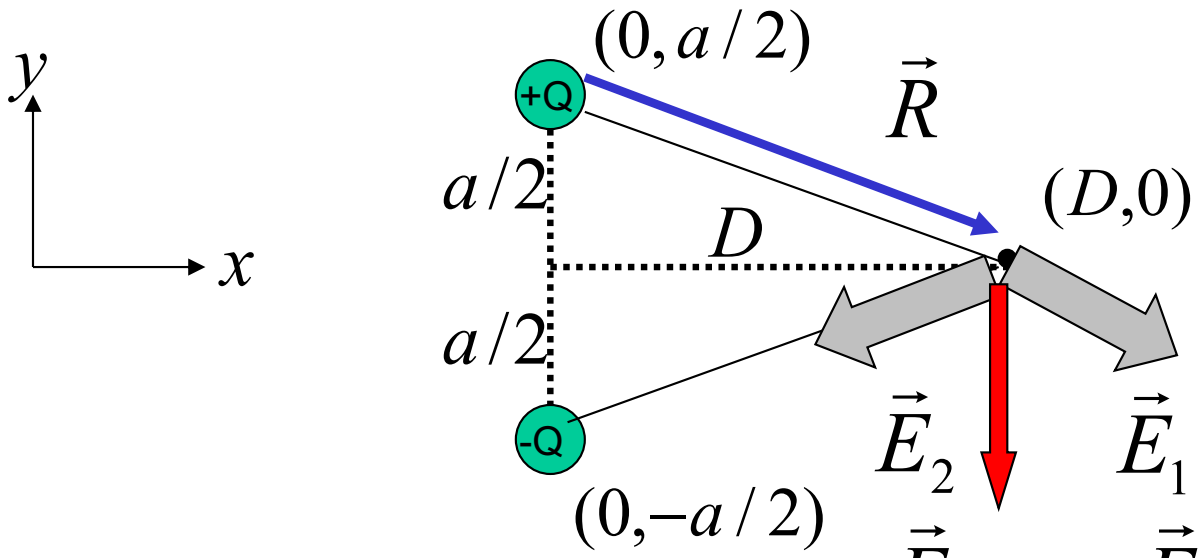
Example: E of Dipole



Tip: Use symmetry to eliminate components that cancel. Here, we only need to calculate the y-component and only for one of the charges (let's say for +Q).

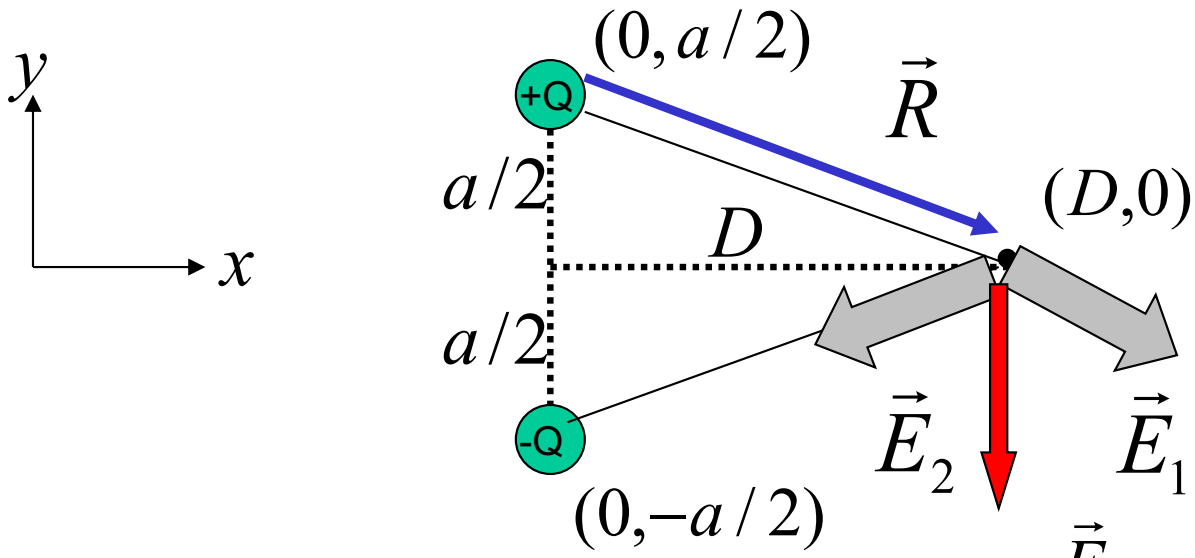
Why?

Example: E of Dipole



$$\vec{E}_{TOT} = \vec{E}_1 + \vec{E}_2 = 2E_{1y}\hat{a}_y$$

Example: E of Dipole

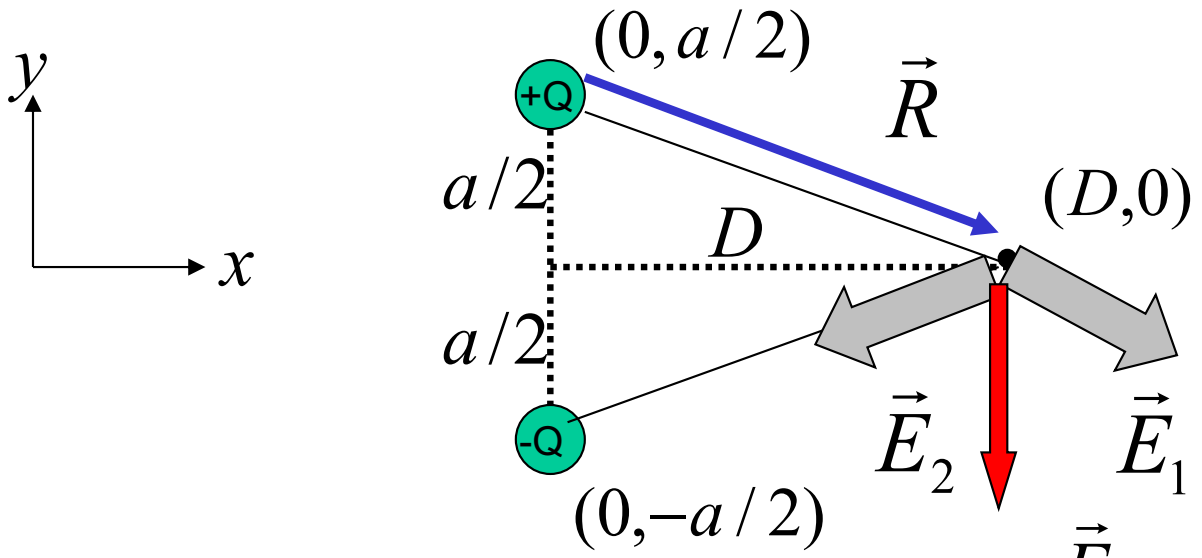


$$\vec{E}_{TOT} = 2E_{1y}\hat{a}_y$$

$$\vec{R}_{from\ +Q\ to\ P} = < x_2 - x_1, y_2 - y_1 > = < D, -a/2 >$$

$$\hat{a}_R = \frac{< D, -a/2 >}{\sqrt{D^2 + a^2/4}} \qquad \vec{E}_1 = \frac{Q}{4\pi\varepsilon_0 R^2}\hat{a}_R$$

Example: E of Dipole



$$\vec{E}_{TOT} = 2E_{1y}\hat{a}_y$$

$$E_{1y} = \frac{Q}{4\pi\epsilon_o R^2}\,\hat{a}_R \cdot \hat{a}_y$$

$$\vec{E}_{TOT} = 2\,\frac{Q}{4\pi\epsilon_o R^2}\left(-\frac{a/2}{R}\right)\hat{a}_y \quad \text{V/m}$$

Patented 5-Step Program for Problem Solving

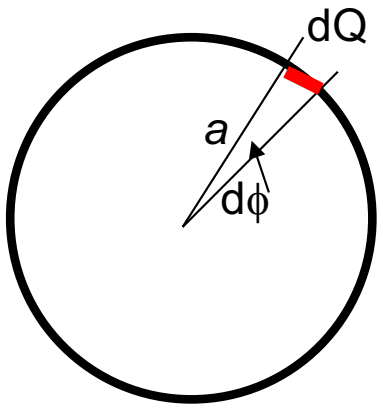
1. MAKE A **LARGE CLEAR** DRAWING
   a. Also draw cross-sections if the problem is in 3D
   b. Pick a coordinate system that is appropriate for the symmetry of the problem
2. Divide charge distributions into tiny pieces dq
3. Find d**E** of one tiny piece
4. Use SYMMETRY to eliminate any components that cancel (i.e. add to ZERO)
5. INTEGRATE to add contribution of ALL the tiny pieces

Example Charge Distributions

- Several discrete points of charge
- Line of charge
- Ring of charge
- Nonuniform lines or rings of charge
- An infinite sheet of charge
- Infinite box of charge
- Spherical surface of charge
- Cylindrical surface of charge
- Spherical volume of charge
- Cylindrical volume of charge

(Review at home)
Example 1: Find Total Charge of a Linear Charge Distribution

Linear Charge λ_0 (C/m) distribution
in a circular loop of radius = *a*

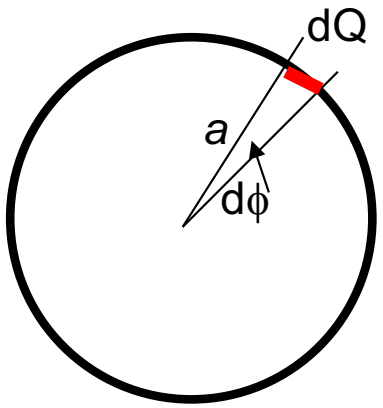


One little piece has a charge (in coulombs)

$$dQ = (\lambda_0)(a\,d\phi)$$

(Review at home)
Example 1: Find Total Charge of a Linear Charge Distribution

Linear Charge λ_0 (C/m) distribution
in a circular loop of radius = *a*



One little piece has a charge (in coulombs)

$$dQ = (\lambda_0)(a\,d\phi)$$

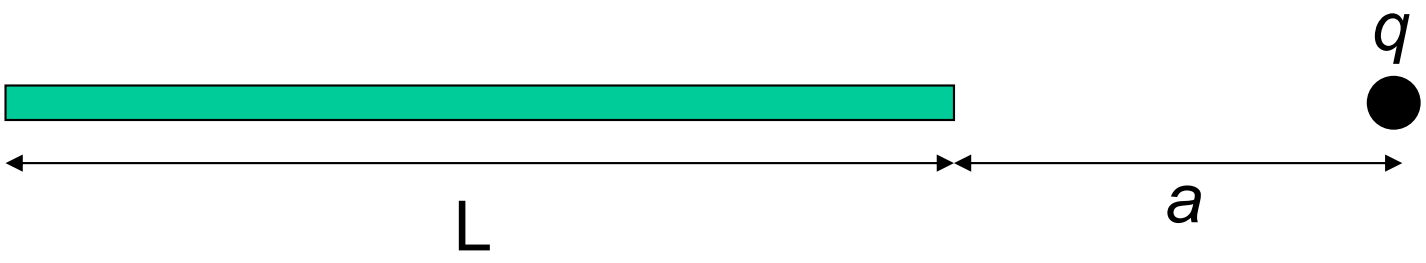
Integrate to get the entire charge of the loop:

$$Q = \int_{\phi=0}^{2\pi} (\lambda_0)(a\,d\phi) = 2\pi\lambda_0 a$$

Units:
Coulombs

(Review at home)

Example 2: F due to line of charge



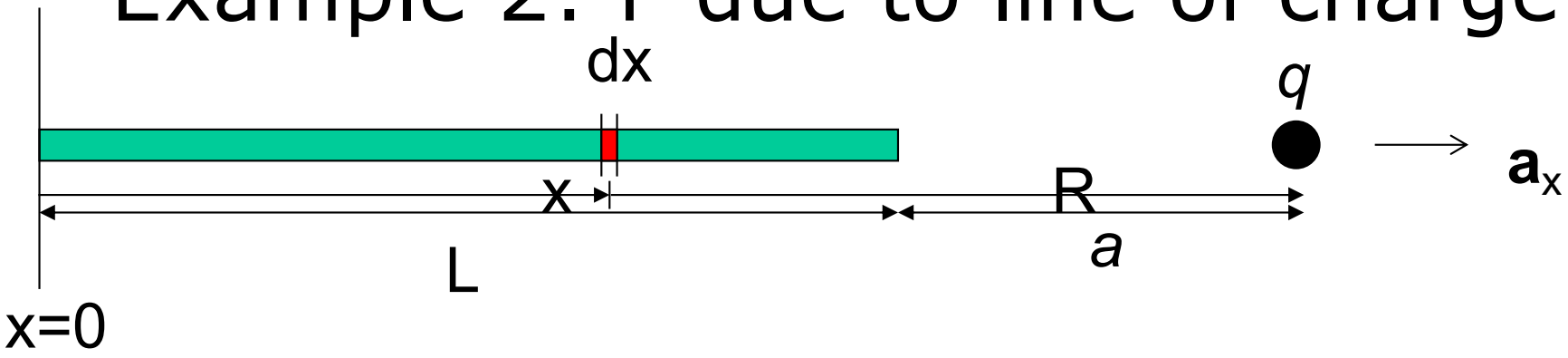
Rod has a TOTAL charge = Q (coul)

So the rod has a charge density $\rho_l = Q/L$ (coul/m)

Find the FORCE exerted by the whole charged rod on the charge q

(Review at home)

Example 2: F due to line of charge



What is the small amount of force, dF, applied by a small sliver of the rod?

Differential force applied to q

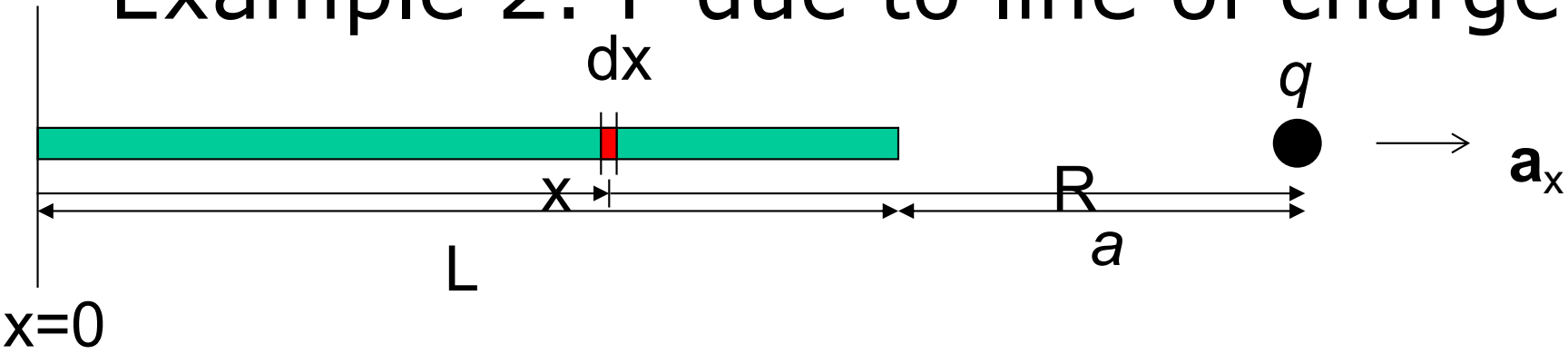
$$d\vec{F} = \frac{\left[\frac{Q}{L}dx\right]}{4\pi\varepsilon_0 R^2} q\hat{a}_x$$

Differential charge in one small sliver (coul)

$$R = (L + a) - x$$

(Review at home)

Example 2: F due to line of charge



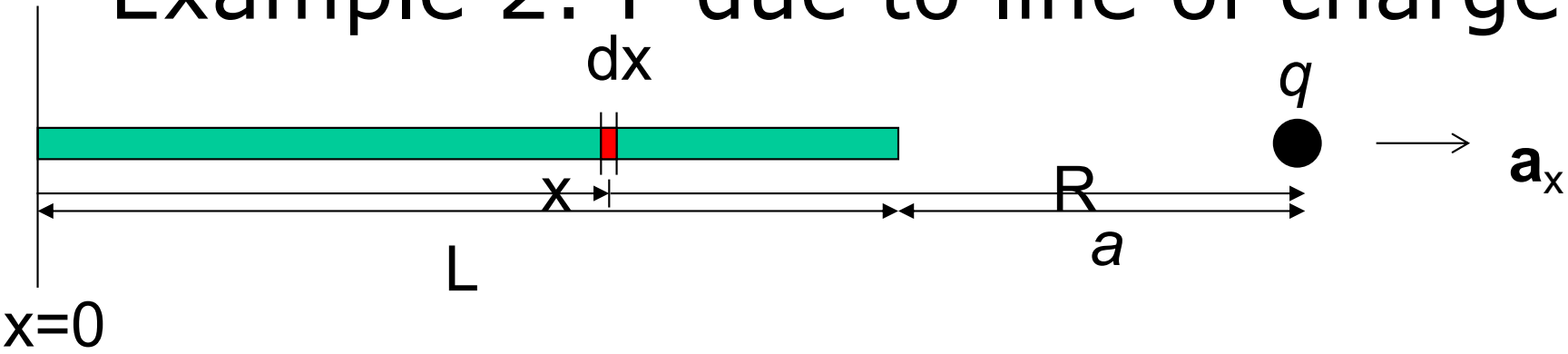
INTEGRATE the force from each part of the rod to obtain the force due to the whole thing:

$$\vec{F} = \int_0^L \frac{qQ}{4\pi\varepsilon_0 L} \frac{dx}{(L+a-x)^2} \hat{a}_x$$

Fortunately, a_x is constant and can be taken outside of the integral. Not so simple in cylindrical/spherical coordinates!

(Review at home)

Example 2: F due to line of charge



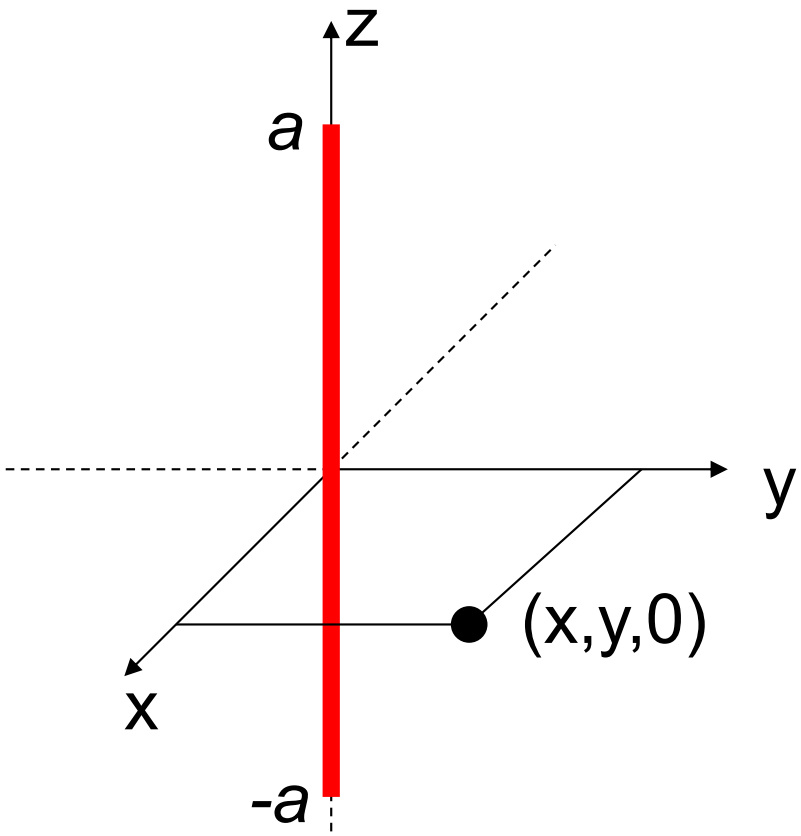
INTEGRATE the force from each part of the rod to obtain the force due to the whole thing:

$$\vec{F} = \frac{qQ}{4\pi\varepsilon_0 L}\hat{a}_x \int_0^L \frac{dx}{(L+a-x)^2} = \frac{qQ}{4\pi\varepsilon_0 L}\hat{a}_x \left. \frac{1}{L+a-x} \right|_0^L$$

$$\vec{F} = \frac{qQ}{4\pi\varepsilon_0 L}\hat{a}_x \left[\frac{1}{a} - \frac{1}{L+a} \right] = \frac{qQ}{4\pi\varepsilon_0 a(L+a)}\hat{a}_x$$

Units: N

Example 3: **E** due to line of charge

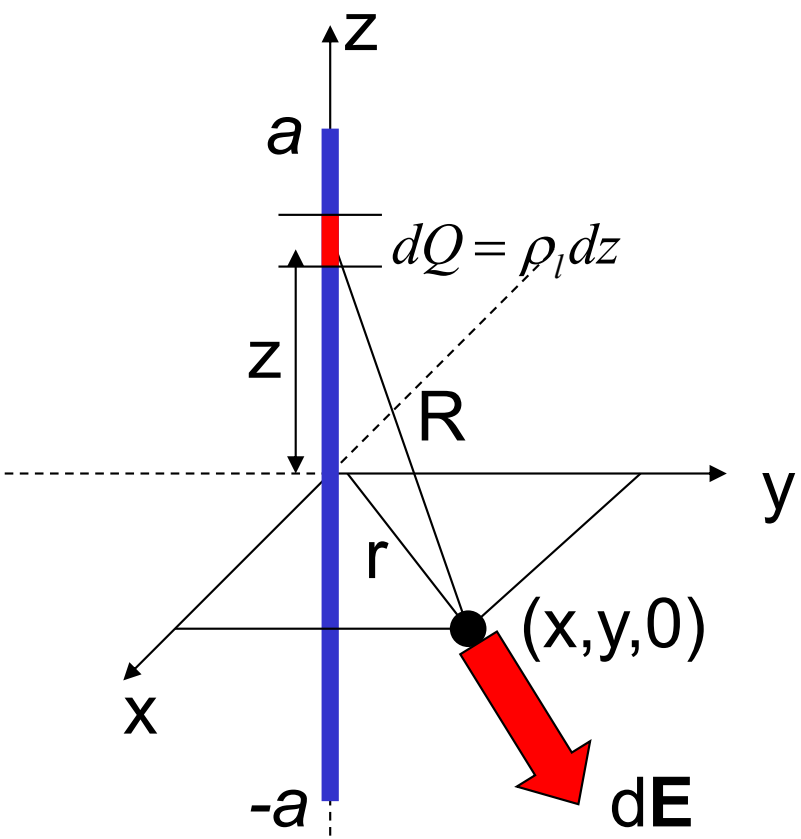


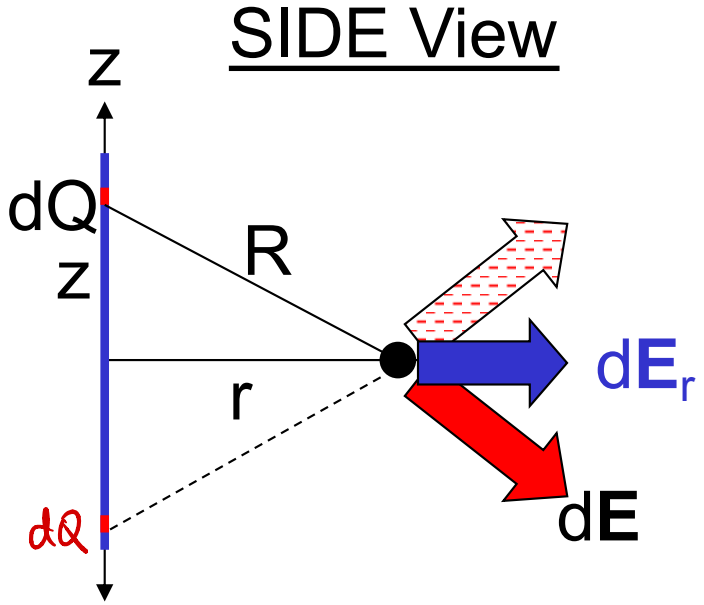
Linear charge distribution
ρ_l (coul/m) from for $-a<z<a$

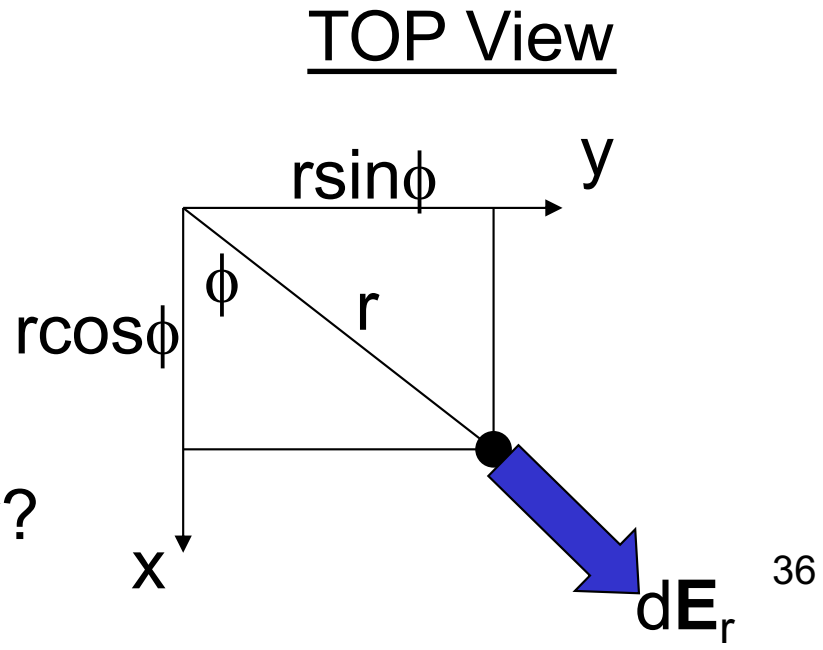
Find **E** at point on xy plane

Next, consider what happens when the line is infinitely long

Example 3: **E** due to line of charge



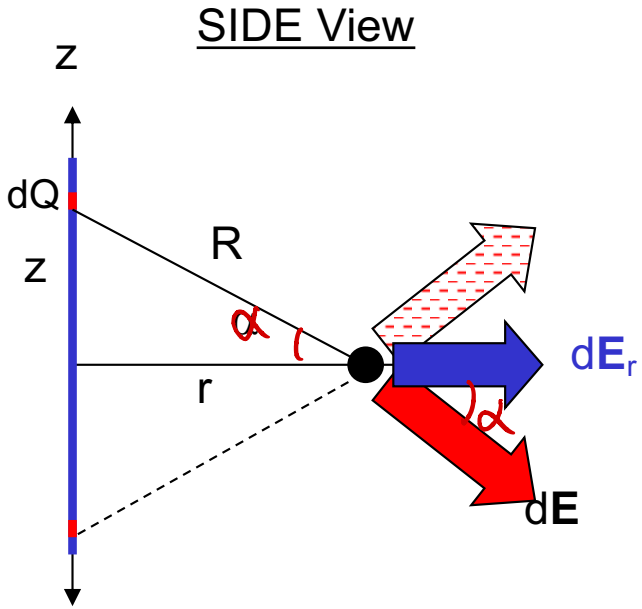




Cylindrical symmetry so use cylindrical coordinates (r,ϕ,z)

$\mathbf{E}_{total}$ will point in a_r direction, why?

Example 3: **E** due to line of charge



$$\overrightarrow{dE} = \frac{dQ}{4\pi\varepsilon_0 R^2}\hat{a}_R$$

$$dE_r = dE\cos(\alpha) = dE\frac{r}{R}$$

$$R = \sqrt{r^2 + z^2}$$

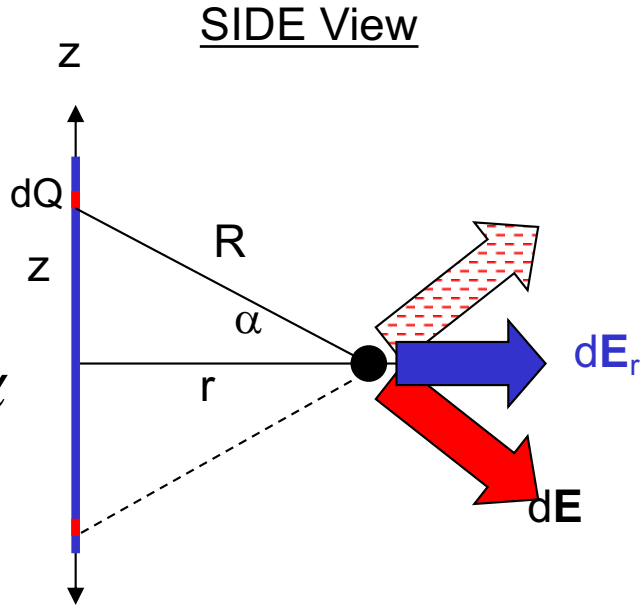
$$E_r = \frac{\rho_l}{4\pi\varepsilon_0}\int_{z=-a}^{a}\frac{r}{\left(r^2+z^2\right)^{3/2}}dz(\hat{a}_r)$$

Slightly difficult to integrate directly

Example 3: **E** due to line of charge

$$E_r = \frac{\rho_l}{4\pi\varepsilon_0}\int_{z=-a}^{a}\frac{r}{\left(r^2+z^2\right)^{3/2}}dz(\hat{a}_r)$$

$$R = \frac{r}{\cos(\alpha)}, \quad z = r\tan(\alpha), \quad dz = r\sec^2(\alpha)d\alpha$$

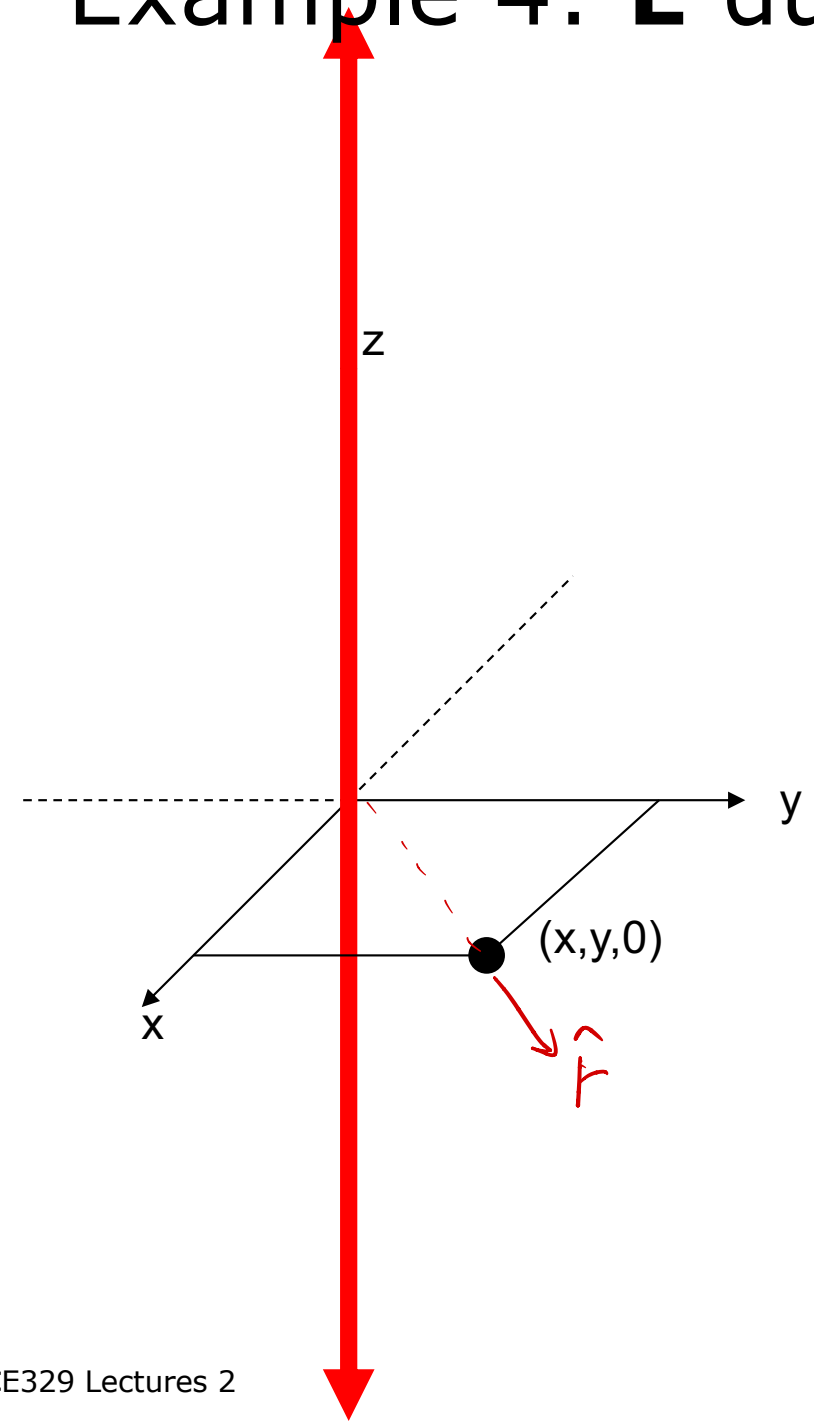


$$E_r = \frac{\rho_l}{4\pi\varepsilon_0}\int_{z=-a}^{a} r\frac{\cos^3(\alpha)}{r^3}(r\sec^2(\alpha)d\alpha)(\hat{a}_r)$$

$$= \frac{\rho_l}{4\pi\varepsilon_0 r}(\hat{a}_r)\int_{\tan(\alpha)=-a/r}^{\tan(\alpha)=+a/r}\cos(\alpha)d\alpha = \frac{\rho_l}{4\pi\varepsilon_0 r}(\hat{a}_r)\sin(\alpha)$$

$$= \frac{\rho_l}{4\pi\varepsilon_0 r}(\hat{a}_r)2\frac{a}{\sqrt{a^2+r^2}}$$

Example 4: **E** due to ∞ line of charge



a >> r so we get

$$\lim_{a\to\infty} E_r = \frac{\rho_l}{4\pi\varepsilon_0 r}(\hat{a}_r)2\lim_{a\to\infty}\frac{a}{\sqrt{a^2+r^2}}$$

$$= \frac{\rho_l}{2\pi\varepsilon_0 r}(\hat{a}_r)$$

$$\boxed{E_r = \frac{\rho_l}{2\pi\varepsilon_0 r}\,\hat{a}_r}$$

point charge

$$\vec{E} = \frac{Q}{4\pi\varepsilon_0 R^2}\hat{R}$$

Field strength now drops off as 1/r,
Not as $1/r^2$ like a point charge

Lecture 2 Summary

- Coulomb's law specifies the electric field of a stationary charge Q at the origins

- For a point charge,

$$\mathbf{E(r)=}$$

- Next class:
  Surface Integrals
  Connecting Coulomb's and Gauss' Law